Machine Learning Prediction of Intensive Care Length of Stay for Lung Cancer Patients

September 2024

Contents

1 Introduction **3**

1.1 Aims and Objectives . . . 3

1.2 Background to Lung Cancer and ICU Treatment . . . 4

1.3 Machine Learning . . . 5

1.3.1 What is Machine Learning? . . . 5

1.3.2 Possible Benefits . . . 5

1.4 The MIMIC Database . . . 6

1.5 Literature Review . . . 7

2 Methods and Materials **10**

2.1 Variable Selection . . . 10

2.2 Database Management . . . 11

2.3 Data Extraction . . . 11

2.4 Covariates . . . 12

2.5 Data Cleaning . . . 14

2.6 Missing Values . . . 15

2.6.1 Missing Data Types . . . 15

2.6.2 Dummy Coding . . . 16

2.6.3 Single Imputation Methods . . . 16

2.6.4 Multiple Imputation and MICE . . . 17

2.7 Prediction Models . . . 18
2.7.1 Linear Regression . . . 18
2.7.2 LASSO and Ridge Regression . . . 19
2.7.3 Random Forest . . . 21
2.8 Evaluation of Prediction Models . . . 22
2.8.1 Performance Measures . . . 22
2.8.2 Internal Validation . . . 22
2.8.3 Nested Cross Validation . . . 23
2.8.4 Cross Validation to Imputation Pipeline . . . 24

3 Results **26**
3.1 Dataset Description . . . 26
3.1.1 Data Extraction . . . 26
3.1.2 ICU Stay Demographics . . . 26
3.2 Missing Data . . . 30
3.3 Imputation . . . 31
3.3.1 Correlation Plot . . . 31
3.3.2 Trace Plots . . . 32
3.3.3 Categorical Missing Values . . . 33
3.4 Model Performance . . . 33

4 Discussion **34**
4.1 Findings . . . 34
4.2 Comparison to the Literature . . . 34
4.3 The Effect of Missing Data . . . 34
4.4 Limitations . . . 34
4.5 Future Recommendations . . . 35

5 Conclusion **35**

Appendices **38**

Abstract

Acknowledgements

1 Introduction

1.1 Aims and Objectives

The main aim of this project is to build accurate and efficient machine learning models using data from the MIMIC dataset to enhance the prediction of the length of stay in the Intensive Care Unit, ICU, for lung cancer patients. The ideal result would be for this to benefit personalised healthcare and resource management in oncology.

The main aims and objectives are the following:

- Aim 1 - Extract the data from the MIMIC dataset to form a final dataset ready to be used to develop predictive models.
  - Objective 1 - Determine the variables for selection
  - Objective 2 - Setup the MIMIC database on PostgresSQL
  - Objective 3 - Extract the data
  - Objective 4 - Clean the data
- Aim 2 - Train various predictive models.
  - Objective 1 - Cross-validate
  - Objective 2 - Impute the missing data
  - Objective 3 - Linear regression model
  - Objective 4 - Penalistation models
  - Objective 5 - Random forest
- Aim 3 - Evaluate the models assessing the effectiveness of each.
  - Objective 1 - Calculate performance metrics
  - Objective 2 - Compare models
  - Objective 3 - Compare findings with other studies

1.2 Background to Lung Cancer and ICU Treatment

Cancer is a disease that involves the uncontrollable multiplication of malignant cells, in some cases forming tumours in organs or tissues. In the US, and many other countries around the world, cancer has become the leading cause of death, and in the US specifically it has been reported that around 1 in 3 people will be diagnosed with any cancer and 1 in 6 of cancer mortality in their lifetimes. This statistic indicates that the volume of patients requiring ICU treatment will be significantly large and therefore the optimisation of length of stay analysis is key to resource management and patient care planning. Lung cancer in particular describes the cancerous cell growth in the lungs and is regarded as a serious health issue for both the patient and healthcare systems worldwide.

According to the official WHO global cancer observatory, lung cancer is the cancer with the highest percentage of mortality and one of the most prevelant. To link this to the statistic previously mentioned, lung cancer will develop in approximately 1 in 17 and be the cause of death in 1 in 28 of Americans. This prevelance and mortality differs when broken down by sex and lung cancer is not the most prevelant or most deadly cancer for women, breast cancer is, however when addressing the whole population lung cancer is the cancer that poses the biggest threat. Focusing on the one cancer that is singled out as the most threatening allows for a more concentrated study that should in theory generate more useful results.

As of Joohyn Park., et al's 2019 report, the mean annual health care expenditure for a lung cancer patient in the US is $35,141. This was significantly higher than the 4 most common cancer types that were included and approximately 8-9 times as high for a cancer free patient, $4484. These numbers point towards the extent of how significant the burden of lung cancer is on the US healthcare system. They also report that lung cancer tends to be diagnosed at more advanced stages so the survival duration is short with a 5-year survival rate estimated at 18.6%. With a higher number of cases diagnosed at advanced stages, intensive care units are expected to be required to administer treatment to patients.

The length of stay in an ICU ward is an important piece of information for clinicians to know and if it can be predicted accurately the efficiency of the intensive care process would improve and so would the quality of care provided. At the time of writing, although some machine learning predictions of ICU length of stay are being researched and perhaps trialled in a limited amount of hospitals, the majority of healthcare systems go by clinical judgement alone. One of the main reasons behind this stems from the lack of understanding regarding how to pair the machine learning

assistance to the clinician's expertise in a way that optimises the blend of predictive knowledge that both provide[1]. The lack of use of these methods in practice provides a rationale for conducting this type of study.

1.3 Machine Learning

1.3.1 What is Machine Learning?

The advancement of computational power and access to large amounts of data has created an opening for the application of statistical techniques to develop machine learning algorithms to produce potentially quite accurate and useful results. Machine learning works similarly to the human brain in that it takes in input information and uses it to build a form of understanding of abstract concepts. Computers will attempt to learn from their input data to identify patterns for use in making predictions or in producing analyses.

There are some things to consider when creating a machine learning model. A model that is trained on data from a specific portion of a population, for example a model that is trained on data from people of a specific country, creates selection bias which renders the model less useful for the rest of the population. So one way to combat this would be to retrain the model on data specific to the population it is intended to be used for. There is also the case of overfitting and underfitting. An underfitted model fails to identify the patterns in the data which may come down to too few variables. An overfitted model is practically the opposite, the model fits too well on the data given therefore performing less well on new data. Adressing these challenges is key to creating and deploying a significantly useful model.

1.3.2 Possible Benefits

Due to factors such as an aging population, medical advancements and staffing shortages, ICUs have now a greater utilisation for severe medical complications such as lung cancer. This raises a few challenges that a robust length of stay prediction could help to solve.

It is imperative for hospital staff to have an understanding of bed availability so that they can manage the influx of new patients effectively. Knowing which beds may shortly become available helps ward managers plan ahead for incoming patients and provides an estimate of how many beds an ICU ward may need.

Length of stay predictions can contribute to a more educated forecast for the ward's running

cost. With this information the hospital can generate a more realistic budget which may reduce the likelihood of excessive spending and underfunding.

At current clinicians do not use machine learning techniques to predict length of stay for lung cancer ICU patients. Their predictions or estimates almost primarily come down to clinical judgement. Therefore estimates may vary depending on a clinicians experience and familiarity to the patient. This prediction is also not an immediate priority for the clinician, they may have the most granular knowledge but they are more focused on administering the best possible treatment to their patients. Therefore a clinically significant machine learning prediction model could help as an aide for hospital staff to make good approximations of the length of stay for an individual patient.

1.4 The MIMIC Database

MIMIC is a large relational database that consists of deidentified data collated from thousands of admissions to the intensive care unit of the Beth Israel Deaconess Medical Centre in Boston, Massachusetts. The acronym MIMIC has stood for “Multiparameter Intelligent Monitoring in Intensive Care” and more recently “Medical Information Mart for Intensive Care”. Some key features of this database are that it is the only freely accessible critical care database of its kind and contains detailed information on patients spanning more than a decade.

The most recent version of the MIMIC database is MIMIC-IV. It includes data collected from January 2008 to December 2022, which is an extension on MIMIC-III which recorded data up to December 2012. Using the updated version should allow for greater relevance and accuracy of the prediction model.

The database is split into four modules, the core data, the hospital data, the ICU data and the emergency department data. For my study I have not included the data from the emergency department module. Within the core there are three tables to store the non-clinical data that can be used for linkage purposes; these are labeled admissions, patients and transfers. The hospital module is the largest, containing nineteen tables. There are tables for administration (services, poe, poe_detail), billing (diagnoses_icd, d_icd_diagnoses, procedures_icd, d_icd_procedures, drgcodes, hcpcevents, d_hcpcs), measurement (microbiologyevents, labevents, dlabitems, omr) and medication (emar, emar_detail, pharmacy, prescriptions). The ICU module includes the d_items, chartevents, datetimeevents, ingredientevents, inputevents, outputevents and procedureevents tables for recording measurement data and the icustays table for linkage.

Data can be accessed after completing a short ethics course set up by the CITI programme

website.[citeMIMICethics]

1.5 Literature Review

This literature review aimed to find relevant peer-reviewed publications to compare with the findings of this study, and to inform on the existing research and the reported results found in this area. The criteria for the initial search were: population of adults over 18, lung cancer ICU studies, length of stay prediction with machine learning, a result of the predictive performance should be stated. A search with these parameters in PubMed found 5 results of which none were relevant. So the search parameters needed to be broadened. The age restriction was dropped and so was the specifity of lung cancer, as in any ICU LOS machine learning prediction study was considered. This search in PubMed produced 697 results of which 13 fit this criteria.

Additionally, 1 publication was found outside of PubMed that fit all of the initial criteria.

Author	Year	Prediction Model	Best Model	Data Source
Ilona W.M.Verburg., et al	2014	OLS regression, OLS regression truncated at 30 days, OLS regression log, GLM Gaussian, GLM poisson, GLM negative binomial, GLM Gamma, Cox PH regression	OLS regression truncated at 30 days: R^2 - 0.208 (0.2001 to 0.215), RMSPE - 5.150 (5.061 to 5.239), MAPE - 3.099 (3.053 to 3.144), BIAS - 0.015 (-0.044 to 0.074)	NICE
Qiuying Chen, et al	2021	Naive Bayes, Linear Regression, Decision Tree, Random Forest, Gradient Boosting Decision Tree	Random Forest: AUC - 0.991 (0.978 to 1.000) and 0.837 (0.766 to 0.908) in the training and validation datasets respectively	Guangdong Provincial Cardiovascu-lar Institute
Khalid Alghatani., et al	2021	Linear Regression, Linear Discriminant Analysis, Random Forest, k-Nearest Neighbour, Support Vector Machine, XGBoost, Multivariate Linear Regression, Support Vector Regression	Random Forest	MIMIC-III

Author	Year	Prediction Model	Best Model	Data Source
Longxiang Su., et al	2021	Logistic Regression, Random Forest, XGBoost	Random Forest: AUC - 0.76	ICU unit of Peking Union Medical College Hospital
Andrea Stieger., et al	2025	Logistic Regression, Random Forest, Stacked Learner, Gradient Boosting Machine, Neural Networks	Stacked Learner: AUROC - 0.94, AUPRC - 0.62	Seoul National University Hospital
Louise Y.Sun., et al	2020	Logistic Regression	Short stay: c-statistic (0.78/ 0.71, derivation/validation), Long stay: c-statistic (0.85/0.78, derivation/validation) This essientially concludes the idea that the logistic regression model may help practictioners	University of Ottawa Heart Institute
Kan Wang., et al	2022	Linear Regression, Naive Bayes, Random Forest, Support Vector Machine, XGBoost	XGBoost: AUC - 0.88	Wuhan Union Hospital
Pepler PT., et al	2012	Poisson Log-linear, Regression Model	Logistic Regression - AUC 0.8507	Vermont Oxford Network
Yucai Hong., et al	2020	Multivariate Regression Model	Multivariate Regression Model: AUC - 0.848	Tertiary Care Hospital in Yueqing City, Zheijiang Province
Geert Meyfroidt., et al	2011	Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Gaussian Forest	Gaussian Processes: RMSE - 0.408, AUC - 0.758	MetaVision PDMS Database
Waqar Aziz., et al	2024	Random Forest, Gradient Boosting	Gradient Boosting: AUC - 0.7243	MIMIC-IV-ED

Author	Year	Prediction Model	Best Model	Data Source
Belal Alsingwali., et al	2022	Random Forest, XGBoost, Logistic Regression	Random Forest: k-fold = 10 mean accuracy - 87.4%	MIMIC-III
Ha Na Cho., et al	2025	Linear Regression, Random Forest, Support Vector Machine, XGBoost, Their own developed model called SurgeryLSTM	SurgeryLSTM: R^2 - 0.86 followed by XGBoost - R^2 - 0.85	University of California Irvine Health
Taghi Khanijev., et al	2025	Logistic Regression, Decision Tree, Random Forest, Neural Networks	Random Forest: AUC - 0.824	MIMIC-IV

Table 1: Caption

From inspection of the table, it seems that the majority of these previous studies focus on building a model to predictions via classification. My intentions however are to build a regression model as I have chosen to define length of stay as a continuous variable rather than a categorical one. Splitting the variable into categories can make results more interpretable and more simple to visualise but in doing so specific information is lost. Losing the specifics may result in a loss of clinical usefulness, affecting the ability to apply more informative bed allocation and staffing needs.

Most of the studies from the table compare multiple machine learning models. The model that appears the most frequently is random forest with XGBoost/gradient boost following close behind. These methods are therefore worth considering as a predictive application within the lung cancer length of stay context. Another takeaway from the table is the frequency of the usage of the MIMIC database. Many use their own locally sourced data but those that don't mostly use MIMIC. Which shows that MIMIC is certainly a viable source data where there is an abscence of data that is privatey collected. Additionally many slightly older studies that use MIMIC will use MIMIC-III. Since MIMIC-IV has now been released, which contains more up to date data than its predecessor and therefore more informative prediction models can be created.

The majority of these studies focus on building models for length of stay prediction in various health related contexts. There is only one study that I have found in my literature search that

particularly focuses on lung cancer ICU length of stay prediction. This study used MIMIC-III data to compare random forest, XGBoost and logistic regression, finding random forest to give the most accurate predictive results. Since there is a scarcity of studies in this area, there is great benefit in further research with the aim of finding a reliable and effective predictive model suitable for use in clinical settings. Their variable selection and feature importance section gives key information on the variables to look at when buildiing a model such as this. They mention the methods of obtaining the variables they have included is a process of first considering clinical significance and then using a process of recursive feature elimination as a way of removing the less useful variables, leaving a refined list that can serve as inspiration.

As the other studies I have found don't focus primarily on lung cancer they may not be as relevant, but variables appearing often will be up for consideration.

The literature research also informs on the complexity of the models, namely the depth of variables used in each study. Most of these studies use many variables in their prediction models. The larger complexity may be favoured for improved performance although with more variables the models become harder to interpret for clinicians.

2 Methods and Materials

2.1 Variable Selection

Variables to use in the prediction models must be determined before any data extraction from the MIMIC-IV database can occur. The importance of choosing relevant variables is significant as this has a strong impact on the performance of the model. Therefore some prior knowledge on the key variables that affect lung cancer is needed, to obtain this it would be useful to draw on medical knowledge. In the absence of first hand medical knowledge some research on the variables used in previous studies must be examined. They mention the methods of obtaining the variables they have included is a process of first considering clinical significance and then using a process of recursive feature elimination as a way of removing the less useful variables. The variables that remained are as listed: age, temperature, platelets, admission (specifically emergency), glucose, chloride, potassium, pulse oximetry, PTT, anion gap, respiratory rate, haemoglobin, troponin, calcium gluconate, haematocrit, tidal volume, diastolic, ondansetron, INR, sodium. Of these pulse oximetry, anion gap, respiratory rate, tidal volume and diastolic were not collected due to lack of data. The variables sex, ethnicity and health insurance were also included with the reason that

they may be clinically significant.

The target variable, length of stay, is to be treated as a continuous variable measured in hours, meaning regression modelling is more suitable than classification in this context.

2.2 Database Management

In order for data to be extracted from the MIMIC database, the data files from physionet should be downloaded and exported into a database schema in pgAdmin 4. From there the database can be queried with SQL commands for data extraction.

To create the schema, i.e. the layout and structure of the database being created, a SQL script is run, specifically tailored for MIMIC-IV, to construct empty tables ready for the data export process. The script was retrieved from a Github repository.[2]

After creating the empty database, the data files will need to be prepared ready for export. Some data files were too large for pgAdmin to export in one process. Therefore the larger data files were split into smaller parts with the use of the Windows powershell command line.[cite command code] Once larger data files had been split they were exported into their respective tables in the schema.

2.3 Data Extraction

Now that the MIMIC database has been setup the data for the determined variables can be extracted.

To obtain the cohort from MIMIC for analysis, the ICD-10 [3]codes corresponding to the diagnosis of lung cancer are needed to identify the patients required for the study. ICD-10 codes represent various patient diagnoses in alphanumeric form, making it easier to record diagnostic data, thereby reducing the likelihood of incorrect entries. The ICD-10 codes that indicate a patient has lung cancer are the C34 codes, identifying a ”Malignant neoplasm of bronchus and lung”, or the C78.0 codes, identifying a ”Secondary malignant neoplasm of lung”. A secondary malignant neoplasm is a cancer that develops as a result of a previous cancer. Along with the ICD-10, MIMIC-IV also uses ICD-9 to indentify health conditions. The ICD-9 codes that correspond to the lung cancer cases that are to be extracted are the 162.9 codes. ICD-10 codelist is the updated version of the ICD-9. ICD-10 was introduced in the middle of the data collection process, ICD-10 was introduced in 2015 and MIMIC-IV was collected between 2008 and 2018. This is the reason both set of codes appear.

ICD Coding System	ICD Code	Condition Description
ICD-9	162.X	Malignant neoplasms of the trachea, bronchus and lung. I.e., cancer that originates in the lung
ICD-10	C34.X	Malignant neoplasms of the bronchus and lung
ICD-10	C78.0	Secondary malignant neoplasm of lung. I.e., lung cancer that develops as a result of a previous cancer

Table 2: Caption

Through pgAdmin 4, SQL queries were written to sift through the database to extract the data to build a dataset containing the desired output laid out in the script. One of these scripts can be used to find data on the lung cancer patients with ICU stays by extracting from the relevant tables.

Similarly, to find data on each variable separate scripts were written each producing datasets extracted from all the data in the database. For example, a SQL query for temperature data would produce a dataset showing every temperature recording in the database. Later, these datasets were to be used to match each admission to their corresponding records, creating a dataset ready for building the machine learning models. Within the SQL scripts there is a set of commands that reduces the data down to only those patients with ICU stays. Since this study does not concern hospital stays without an ICU admission, these commands were utilised so that each of the datasets included only admissions with ICU stays. The data required is located by finding the specific label that identifies the data needed for extraction, for example, for finding temperature data, the WHERE command is used to find data labelled in part with ”temp” in the labels table.

Should there be multiple values before the first ICU admission the value recorded closest to the admission is extracted.

The datasets, now compiled, were saved into CSV files and imported into one R file. After adjustment, as outlined in the data cleaning section, the datasets should all be collated into one.

2.4 Covariates

To predict length of stay the model will need relevant data as its input. Data from variables of interest are collected, these are refered to as the covariates. The table below gives the list of covariates

to extract from MIMIC, selected for their use in relevant literature identified in the literature review section and for their relevance.

Variable Name	Unit of Measure	Data Type	Description
Age	years	Continuous	The number of full years the patient has lived
Temperature	℃	Continuous	The closest temperature recording before admission into the ICU
Platelets	ku/L	Continuous	Cell fragments that allow the blood to clot and injuries to heal
Chloride	mEq/L	Continuous	An essential electrolyte that works in a capacity to maintain pH balance, help digestion and cell function among other uses
Sodium	mEq/L	Continuous	Sodium works together with chloride in many bodily functions. It is also a major nutirent involved in water retention and blood pressure regulation
Potassium	mEq/L	Continuous	Electrolyte that is useful to lower blood pressure by removing excess sodium. It also promotes bone density and healthy muscle function
Glucose	mg/dL	Continuous	Responsible for energy production and brain function
Haemoglobin	g/dL	Continuous	Important for the transferral of oxygen from the lungs to the rest of the body and takes carbon dioxide from cells to the lungs for it to be exhaled
Haematocrit	%	Continuous	This is the percentage of the amount of red blood cells in the blood
Partial Thromboplas-tin Time (PTT)	seconds	Continuous	The time taken for blood to clot
Troponin	ng/L	Continuous	Troponin helps the heart muscle contract and relax

Variable Name	Unit of Measure	Data Type	Description
Calcium Gluconate and Ondansetron	no unit	Binary	Calcium Gluconate and Ondansetron are both medication. Ondansetron aims to prevent nausea and vomiting caused by cancer treatments or surgery and Calcium Gluconate is used in cases where calcium is too low, potassium is too high and/or magnesium is too high
Ethnicity	no unit	Categorical	Categories of universally recognised ethnic groups
Sex	no unit	Binary	Two binary categories, male and female, as recognised in the MIMIC-IV database
Admission Type	no unit	Categorical	Categories that represent the type of admission for the patient
Health Insurance	no unit	Categorical	The type of health insurance the patient has

Table 3: Caption

For each continuous covariate the data value that is used in the prediction is the value that is taken closest to the admission time.

2.5 Data Cleaning

SQL and R programming [Ref] were used to handle data cleaning.

The ICU length of stay value is defined by the length of stay of the first ICU stays and the duration of any readmissions that occur within an hour of prior discharge. An admission occuring outside a day between prior discharge and the start of admission is considered a seperate admission in the data. An extra variable is created, indicating the number of previous ICU stays that have occured within each admission.

Some of the categorical variables, extracted from the dataset, had many specific levels with some including an insufficiently low amount of individuals. For example the ethnicity variable was split down into many groups, some caputuring only a few individuals. Each group may be incor-

porated into a more broad ethnic umbrellas. For example, white - eastern/european and white - russian are recognised as different ethnic groups in MIMIC, but both can be included within a white ethnicity category. Individuals from ethnic groups that don't fit into a specific category are placed in a category labelled "Other". Admission type categories were also aggregated.

Any missing data in a categorical variable was placed into its own missing data group.

2.6 Missing Values

Missing values in the data act as a roadblock to meaningful analysis. Many machine learning models require the input data to be complete for them to work, and their performance is greatly improved when the missingness is remedied so that the resulting study population matches the true population distribution as closely as possible[4].

2.6.1 Missing Data Types

There are three distinct types of missing data, missing at random (MAR), missing not at random (MNAR) and; missing completely at random (MCAR). Data is considered MCAR when the missingness of the data is random. This differs from MNAR which describes missing data that depends on the context behind the missing value. For example people are less likely to disclose their substance use due to societal judgement and/or its legality. And finally MAR is dependent on data from other variables[5]. The importance of these distinctions lie in the determination of the choice of imputation or deletion method of dealing with this missingness.

Dealing with missing data can involve the complete case analysis omission of records or imputation. Although omission is the simpler method, the resulting model is subject to bias as there due to the possibility for certain groups to have missing data, of the type MAR or MNAR, in some variables. Therefore, omission of these data could potentially lead to a misrepresentative study population which in turn leads to the aforementioned bias in the prediction models[6].

There are some forms of imputation that are a much more valid approach for minimising bias. Imputation is, in simplistic terms, the filling in of missing data, as an approximation, to provide a full dataset for analysis. The imputed data is based on the statistical characteristics of the observed data such as its distribution.

2.6.2 Dummy Coding

Categorical data are not handled well by machine learning models as they are designed to make predictions from data in numerical form. This is where dummy coding comes in. The idea, to put it briefly, is to assign binary values as an indication as to which category the record falls into. For example if a categorical variable has four categories then three separate columns would be created. Let's say a patients admission type is urgent, then the value in the urgent admission column would be 1 and in all other admission type columns the value would be 0. category in that variable is urgent admission then every admission record included in this category would receive a value of 1 and the others a value of 0. As would be the same for all but one of the other categories in that variable. The one variable this is not the same for is the variable that isn't given its own column. To determine if the patient is in this categorical group each of the other categories, which all have a column, are set to 0. In this way dummy coding differentiates from a similar method of converting categorical data to numerical known as one-hot encoding.

Dummy coding is also helpful for eliminating the need to impute missing categorical data. A missing data category can created for use in the coding process.

2.6.3 Single Imputation Methods

Single imputation is a basic imputation method that imputes the missing data with single estimated values. It is most important to note that these methods only impute with one iteration. One way to singularly impute the missing data is to replace it with a summary statistic, such as the mean or median for quantitative variables or mode for qualitative variables[7]. For datasets with a significant amount of missing data this imputation method is less suitable and not recommended for producing robust estimates as the variance of the resulting imputed distribution would almost certainly be a poor estimation of the variance of the true distribution.

Other methods offer an improvement to this simplistic form of imputation. The hot deck method matches individuals based on the similarity of their characteristics[8], and uses this information to impute the missing data in the dataset. The individual with the missing data is labelled the recipient and the individual with the data used to impute the missing value is labelled the donor. How the donor is selected is dictated by the type of hot deck imputation [9]. As opposed to imputing all missing data with the same value, methods such as hot deck or regression give imputations that differ for each missing value depending on the characteristics of their records.

Regression imputation uses data from other variables and imputes with regression techniques, such as those used to predict the length of stay in this study. It is worth noting that both regression and hot deck require the variables used in the imputation to be fully-observed. Therefore, these methods are less effective for imputing datasets that contain a lot of missing data.

Single imputation is not the optimal method when missingness in the dataset is more prevalent and therefore advanced methods should be considered.

2.6.4 Multiple Imputation and MICE

Multiple imputation improves on single imputation in several ways with the use of multiple imputed values for the missing data and allowing for every variable, including those with missing data, to be used in the imputation process[10]. MI works without modification if the missing data included in the dataset are either MAR or MCAR. For data that is MNAR sometimes with the appropriate adjustments to the data multiple imputation methods are suitable.

A common MI method is known as multivariate imputation by chained equations (MICE) (2). This method imputes every variable using data from the other variables as predictive inputs the defined model whether it be regression, random forest or some other kind.

The MICE method involves an iteration process of imputation through using the data from the other variables as predictors. The ideal end of the process comes when each variable's, with missing data to start with, distribution converges to a point of insignificant difference between iterations[11].

As stated previously, the method of imputation within the MICE process can be changed, especially for the goal of matching the distribution as closely as possible. Simple methods include imputing each time with the mean of the observed values, although technically this is just an iteration of single imputation. Others make use of the benefit that multiple imputation, and MICE in particular, brings. Within the MICE R function, methods such as predictive mean matching, (PMM), Bayesian and random forest can be selected. This paper will solely focus on the PMM method but conducting a sensitivity analysis on each of the methods should determine the most useful.

PMM involves a process of regression and nearest neighbour selection to impute its values. Its first iteration takes the observed values to build a regression model, via linear regression, to impute the missing values. The next step is taking k nearest neighbours. Each imputed data point has k values from the original observed data that are closest to it. The idea is to then take a random value and replace the value from the imputation via regression. This process is repeated in each

iteration, where, in this instance, every value including those observed is replaced[12].

In this project, I assume data are MAR .

2.7 Prediction Models

The primary outcome of interest in this study is the metric evaluating the lung cancer patients' length of stay in the ICU. Therefore, experimenting with machine learning prediction models such as linear regression, penalisation methods and random forests will help identify a suitable and optimal model.

2.7.1 Linear Regression

Linear regression is, compared to other widely known machine learning prediction methods, somewhat simple and easy to understand. It is often used as the first step in many attempts to apply machine learning techniques for prediction purposes. "This paper" shows how a simple linear model can be more effective than other more complex models, with the example of a study involving breast cancer diagnosis prediction[13]. All of which suggests that this method should be evaluated first.

The method itself comes down to the relationship between a dependent variable to be predicted and the independent variables used in the relationship as predictors[14]. Included in this equation is a term as an indicator for the noise evident in the model, in fact one of the main aims in improving the effectiveness of the model is the minimisation of this term. This is referred to as ordinary least squares regression or OLS when the error terms have constant variance.

A univariable version of this model includes only one independent variable for prediction and can be described with a simple equation shown below.

$$y = \beta_0 + \beta_1 x + \epsilon$$

Here, y is the predicted value of the dependent variable. In the case of this report the length of stay in the ICU within a lung cancer patient's admission. β_0 refers to the y intercept given by the equation, which can be visualised by plotting the linear relationship of the model. β_1 is the coefficient of the variable x and is symbolic of the effect the variable has on y. Finally, ϵ represents the error term that indicates the noise in the prediction.

Since this prediction model is likely to include several independent variables, it would be more apt to model using the equation for multivariable linear regression. The equation's structure is very similar to the equation for univariate regression but with the added terms to represent the effect of the extra variables on, in this case, the lengths of stay. The equation is as follows.

$$y = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p + \epsilon$$

The number of terms in multivariate regression is represented as the letter p in the equation.

Since the key goal in improving a linear regression model is minimising the error term the equation that the computer code seeks uses to do this is a rearranged version of the earlier equation which is written to make predictions for the depedent variable y. The rearranged equation simply puts the error term in the place of y. Since the sum of the residuals, a name for the error term values, is always zero minimising the sum of the residuals would not make any sense. Therefore the sum of the squared residuals must be observed to evaluate the model. The equation is called the residual sum of squares and is shown below.

$$RSS = \sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2$$

With i representing an individual record summing to the sample size of n and j representing a predictor variable among p predictor variables. So the function to build the model within the code would find the β values to minimise this equation, theseβ values can then be used in the original equation to predict the dependent y variable.

2.7.2 LASSO and Ridge Regression

LASSO and ridge regression models introduce a β coefficient penalisation term which minimises their value towards zero to reduce overfitting in the model. When the model is overfitting it is trained too well to make predictions on the training data. This is a problem because any new data used to make predictions would produce predictions that may be far from equivalent to the actual values. These two regression models both introduce a λ term which needs to be tuned carefully to find a model that reduces overfitting, improves prediction and does this without creating an overly complex model. The equation for these two models is very similar to the equation for the OLS. For ridge regression the equation is as follows.

$$RSS + Penalty = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}\beta_j^2$$

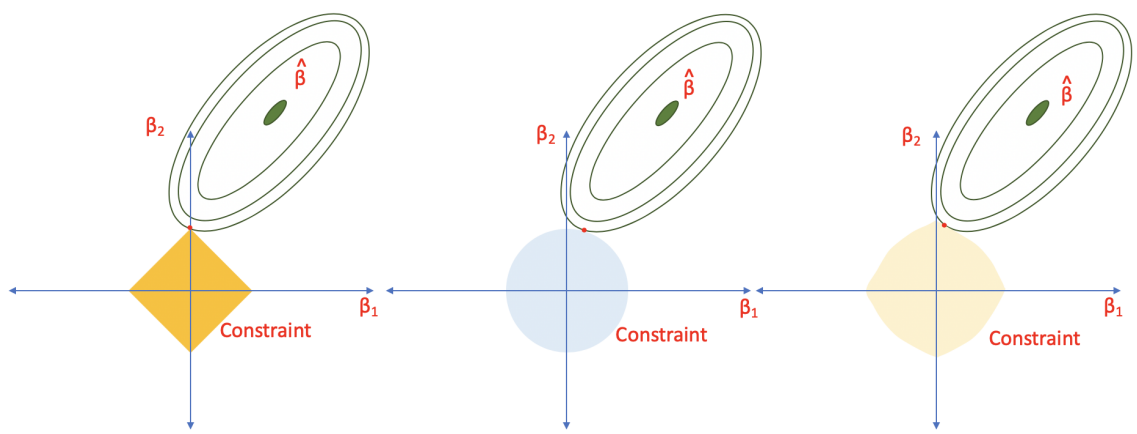
And the LASSO equation is written below.

$$RSS + Penalty = \sum_{i=1}^{n}(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij})^2 + \lambda \sum_{j=1}^{p}|\beta|$$

In both these equations $\lambda \in [0, \infty)$.

Here we see the difference in the two models and how they are both linked to OLS. Essentially when the value of λ is zero there is no penality so you are left with the minimisation of the RSS alone and an equation for the OLS with the β values being the same as they are in this model.

Plotting the contours of the error and the constraint regions [datasklr cite] help to visualise how the β coefficients are pushed to zero with these methods.



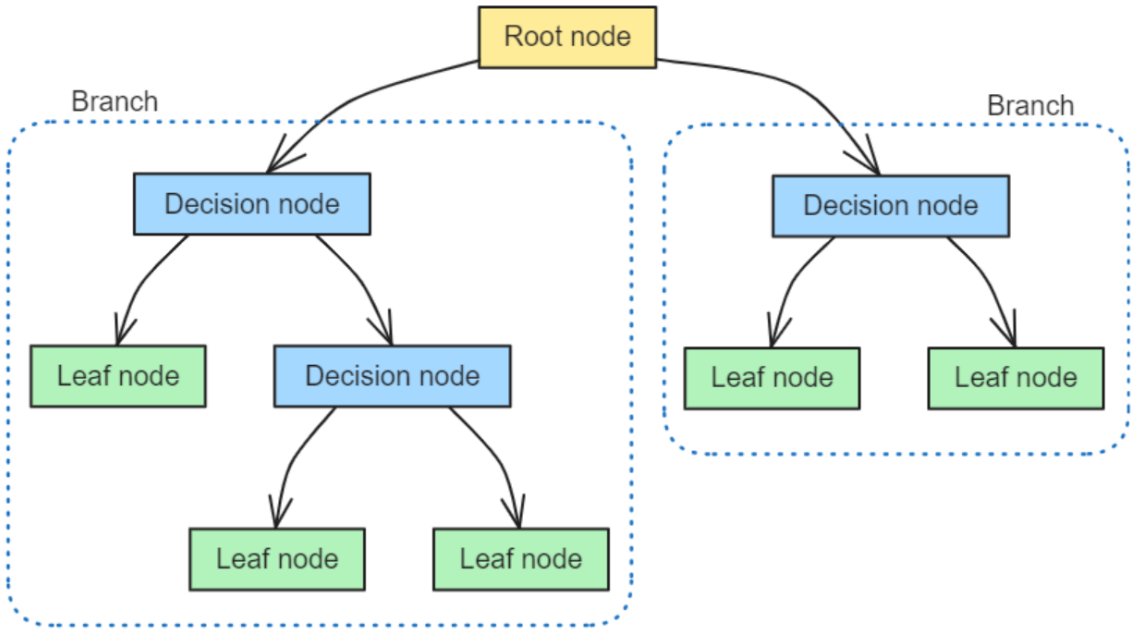
So what can be seen here from the three penalisation model graphs are three differing shapes for the constraint regions. The ellipses represent the contours of the RSS values away from the OLS. The constraint regions will increase in size as $\lambda \to 0$ which would bring the β values to their OLS values. However, if λ does not equal zero then the constraint shrinks along with the β values as they are evaluated at the point where the RSS + penalty contours intersect with the constraint region. The reason why LASSO pushes some coefficients to zero is shown in these plots. Because of its diamond shape there is a possibility that the contours intersect with the constraint region on the axes. This is not possible with ridge regression since in OLS no coefficients have zero values

and therefore the contours will not intersect the circular shape at the axes. The third plot from the right shows the constraint region for an elastic net which is the middle ground between LASSO and ridge regression. Here the coefficients may get pushed quite close to zero but will never reach it.

The plots show only a relationship between a dependent variable and two independent variables, hence the ability to plot this on an x-y plane. For more than two independent variables the plots get more complex and include further dimensions so they are very tricky to display on paper

2.7.3 Random Forest

To explain the random forest method, the concept of a classification and regression decision tree (CART) must be established first[15]. At the beginning of constructing a decision tree 100% of the data constitutes what is labelled the root node[16]. The root is split into two groups/child nodes, according to an attribute of one independent variable[17], so that the overall mean squared value is optimally minimised[18] . The splitting continues until a stopping point is reached. This may be because the mean squared error minimisation offers insignificant improvements or when the process has concluded in a predetermined maximum number of splits. As a result the data in the terminal nodes should have a significant level of homogeniety. Each terminal node includes a subset of the overall population wherein the mean for the values of the target variable constitutes their predicted value [19]. Therefore, when evaluating the model, the test data is fed into the tree and their predictions are determined according to which terminal node they fall into. The evaluation metric can then be calculated using these predictions in comparison to the actual values.



A single decision tree achieves low training error due to its high flexibility, meaning that the model is more likely to fit to noise in the data. Thereby the model is prone to overfitting. The lack of generalisability on unseen data, especially when making predictions on the test set, is a major drawback for using a single decision tree as a prediction model. To mitigate overfitting and reduce variance, multiple decision trees can be aggregated within an ensemble framework. One such ensemble method, that is widely used, is the random forest algorithm[20]. The final predictions that are obtained from a random forest model are averages from the predictions made by multiple independent, parallel decision trees[21]. The data used in each decision tree is sampled with replacement, a method known as bagging (bootstrap aggregation). Since each tree sees different data, the overall predictions are less affected by noise and variable errors which cancel out when averaging the predictions resulting in lower variance. Another feature of the algorithm is random variable selection. At each split only a subset of variables are considered. This promotes diversity among the trees. A random forest model is therefore more robust.

2.8 Evaluation of Prediction Models

It is important to consider the validity of prediction models. There are several methods that are designed to check the generalisability of how well models fit to new data which can be assessed both using either internal or external validation[22]. A prediction model needs to go through this process of validation so that it can be compared with other models and also it assists in determining whether the model may be suitable for use in practice.

2.8.1 Performance Measures

2.8.2 Internal Validation

A common method of internal validation is k-fold cross-validation[23]. This involves splitting the dataset into k folds, usually k is set to 5 or 10, one split is used as the test dataset and the others used to train the model. In this way, all the data is used both to train and test. The performance metrics in each fold can be evaluated using the predictions and actual values for the test set. k-fold cross-validation is an improvement on the simpler method of split-sampling, which is essentially a cruder version of the method. Split-sampling could be thought of as equivalent to a 1-fold cross-validation just without rotation on different folds as the data is split into only one training set and one test set. The reduced use of the complete data to both train and test, as is the case in cross

validation, introduces a higher level of variability and bias in estimating the performance of the model[24]. This is why the cross validation technique is highly favoured in comparison.

The validation method used in this study will be k-fold cross-validation due to its managable computational intensity and its ability to work well with larger datasets.

Other validation methods may be used but they come with significant downsides. For example the jackknife and bootstrap methods are not suitable because they are much less effective for larger datasets due to their high computational complexity and therefore computational cost. Bootstrap can introduce bias and it is not reasonable to use for a length of stay prediction as the length of stay is often highly skewed to the right which bootsrapping doesn't handle well. Jackknife mutes the potential for exploring nonlinear variables as for this it is unstable.

The best way to validate a machine learning prediction model is to validate externally. A model validated only on the data included in the original dataset still carries a lot of optimism and may fit poorly. Evaluation on relevant data obtained in new settings gives a better estimate of how generalisable the model is to a diverse range of populations. The more external validations performed with a more varied array of data provide a greater picture of the generalisability. This is important because if the model is applied to populations with demographics dissimilar to those represented in the training or testing data, the likelihood of inaccurate predictions increases. Inaccurate predictions lead to inefficiencies that reduce the model's usefulness.

2.8.3 Nested Cross Validation

A simple k-fold cross-validation is prone to some bias. A way to reduce this bias is to use a process of cross-validation within each fold, which is a method commonly known as nested cross-validation. A visualisation of the process is given in figure 8 of this paper[25], it shows how the train dataset is split for cross-validation. The bias that arises in standard cross-validation stems from the fact that the test data used in evaluation will also be used in hyperparameter tuning, essentially saying the test set has been peeked at in the process. The bias affects the accuracy of the performance evaluation, so it should be reduced as much as possible. The nested approach minimises the bias significantly by tuning the hyperparameters solely on the train datasets. These points echo the understanding that, Krstajic et al [26], present. Of course the downside of nested cross-validation is the increased rounds of training and testing which leads to a high level of computational complexity that may dissuade its usage depending on the computational power at hand.

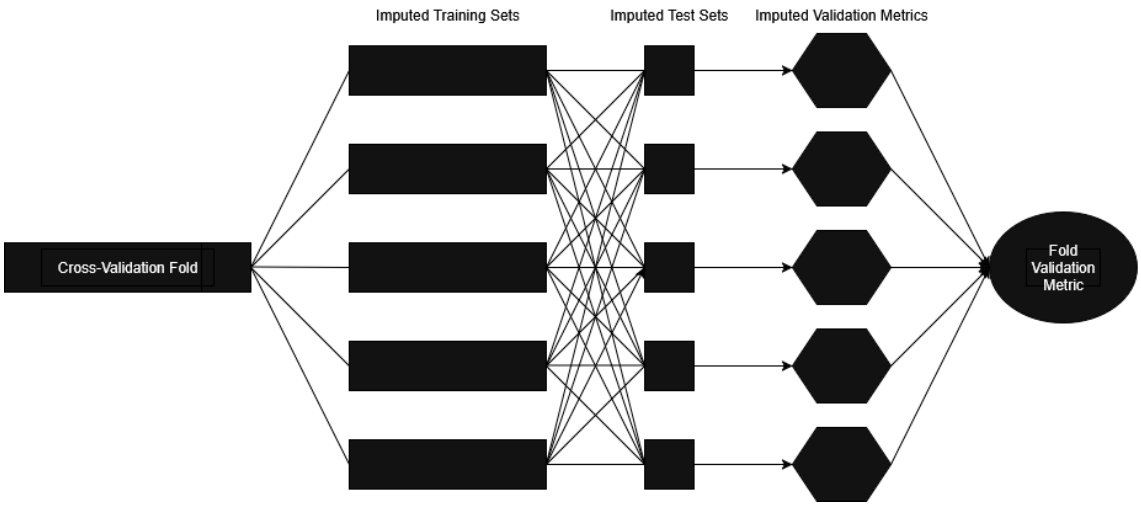
2.8.4 Cross Validation to Imputation Pipeline

To evaluate the performance of the models it's important to consider the order of pre-processing (imputation) and cross validation. The order is important for reducing data leakage[27], which introduces optimistically biased predictions of the observed data causing the reduction of the suitability for use of the model to make predictions on unseen data.

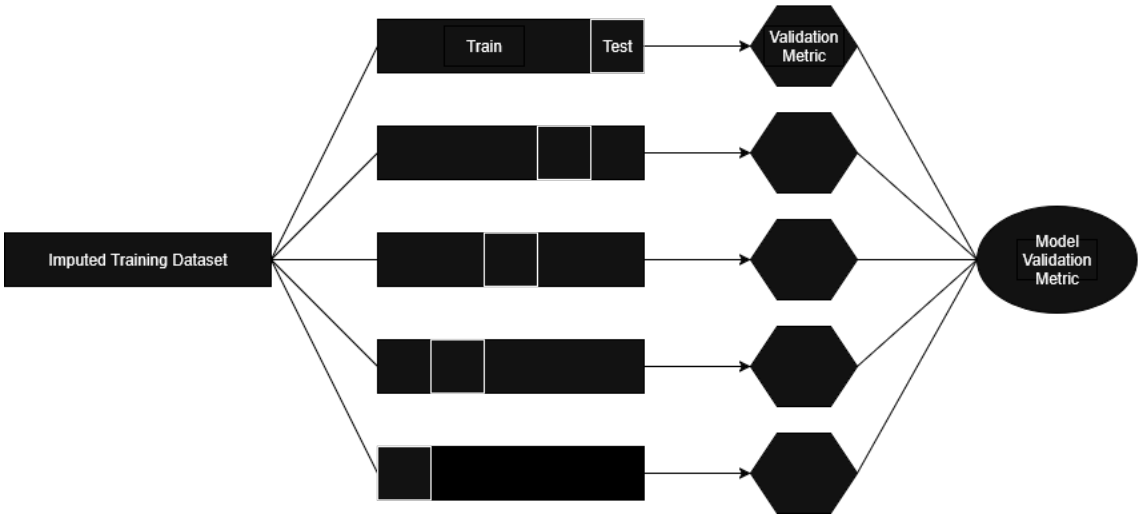
By imputing first, the data is imputed only once, so any data used in validation affects the way in which the missing data is imputed thereby leading to a significant amount of data leakage. In comparison by using an imputation within each cross validation fold approach the data leakage is nullified because the data from the test dataset never affects the imputation of the data used to train the model. So in terms of avoiding data leakage the order of imputation within the cross validation would be preferred; the evaluation that imputation within cross validation outputs is less optimistically bias and closer to the true evaluation to the model's ability to predict accurately on unseen data.

Imputation before cross validation before cross validation results in a lower variability whereas imputation within cross validation results in a higher variability. In a validation process there a several sources that affect the level of variance. With the core structure of the two approaches in question the only source of variance that separates the two is how the imputation is applied. Imputing the whole dataset before it's split up into folds means that the variance gained from imputation is only appended to the total one time. In contrast, for the opposing approach, imputation is conducted within each fold and therefore the total variance will be higher in this scenario. A lower variability in the prediction model leads to more stable predictions as the confidence interval is low. In the case of ICU length of stay, a low variability would therfore predict stays of similar lengths with very little room for outliers. High variability introduces a greater amount of noise and is much more well accounted for picking up on outliers. This is beneficial for predicting the nature of ICU length of stay as there can be stays that last condsiderably longer than predicted unless high variability is accounted for. In some instances low variability would be the more suitable option but due to the data leakage and the context the order of imputation within cross validation with its high variability is favoured.

So with the decision made that the order of imputation within cross validation as the more suitable choice a visualisation is useful to properly show how this works.



This visualisation shows how to implement the MICE imputation process within one cross validation fold to produce a value of the validation metric. The cross validation fold with missing data is imputed five separate times. The train set is imputed first, then the corresponding test set is imputed by using the imputer devepoloped when imputing the train set. Since the order of the validation is imputation within cross validation each imputed set of train and test sets should be validated against each other. This is shown in the figure by the various arrows representing the many-to-many validation to produce five validation metrics values, these are averaged to obtain the value of the validation metric for this fold. This value is used amongst the other values of the fold validation folds to produce the final validation value of the final model.



This figure visualises the inner loop of nested cross-validation which is the result of the decision to add a layer of cross validation within each fold. As shown, each imputed train dataset is taken and split into k-folds, just as in the outer loop. The process then plays out as a standard k-fold cross

validation as is detailed above. This repeats in an iterative fashion as it cycles through values of the hyperparamer, which this inner loop attempts to tune.

3 Results

3.1 Dataset Description

3.1.1 Data Extraction

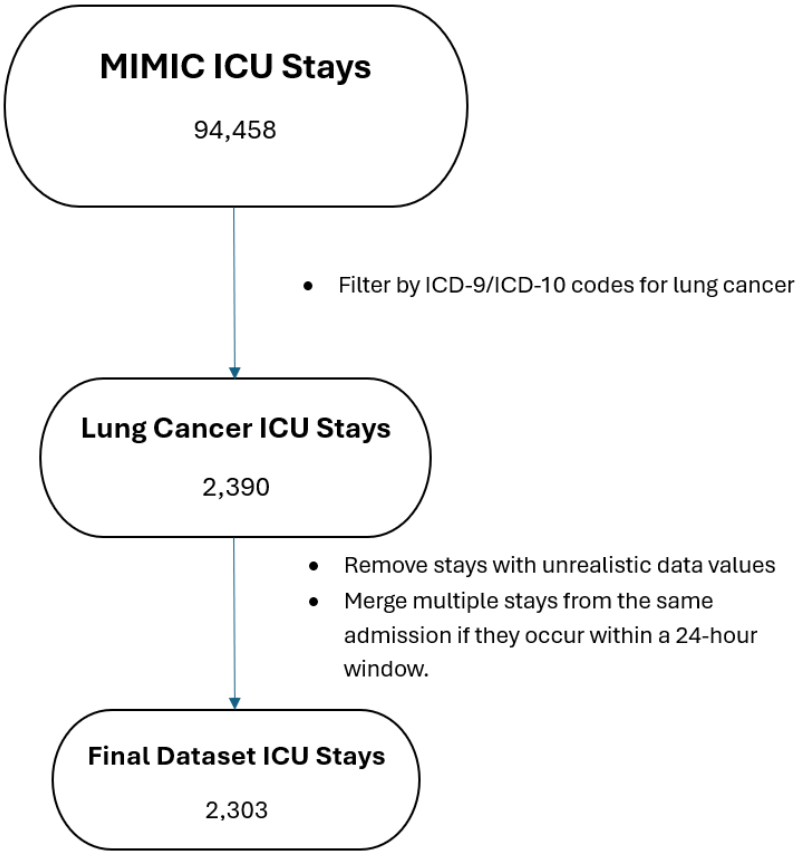
The MIMIC-IV dataset provides data on 364,627 unique individuals who give 546,028 hospitalisations with 94,458 unique ICU stays. To identify the lung cancer ICU stays the data in the ICU stays table must be sifted through by the ICD-9 and ICD-10 codes that identify lung cancer. Data is removed if it is outside of the normal range in any of the variables. After doing this the number of ICU stays for patients with lung cancer comes to a value of 2,390, sourced from 2,115 unique admissions.

An admission may have multiple ICU stays recognised within the dataset. If two concurrent ICU stays are less than or equal to 24 hours apart their lengths of stay are added together and treated as one stay. If this is not the case they will be treated as two separate ICU stays. A column is created to differentiate the ICU stays from the same admission marking the number of previous stays. This reduction leaves the dataset with 2,303 ICU stays.

The flowchart in figure ... below shows how the ICU stays, used in the prediction models, are extracted from the MIMIC-IV dataset.

3.1.2 ICU Stay Demographics

Among the 2,303 ICU stays there is a slight favour of males, 54%, to females, 46%. The most populous age range is the 61-70 year old group, 33.4%, followed by the 71-80 group, 29.0%. The 18-40 group is markedly small, 0.6%, given it's wide range. The majority of patients' ICU stays, included in this dataset, are of white ethnicity, 72.6%, and the majority of patients are in hospital with medicare insurance , 61.6%. Over half of the stays, 55.5% were from the result of emergency admissions. More detailed statistics is given in table 1.



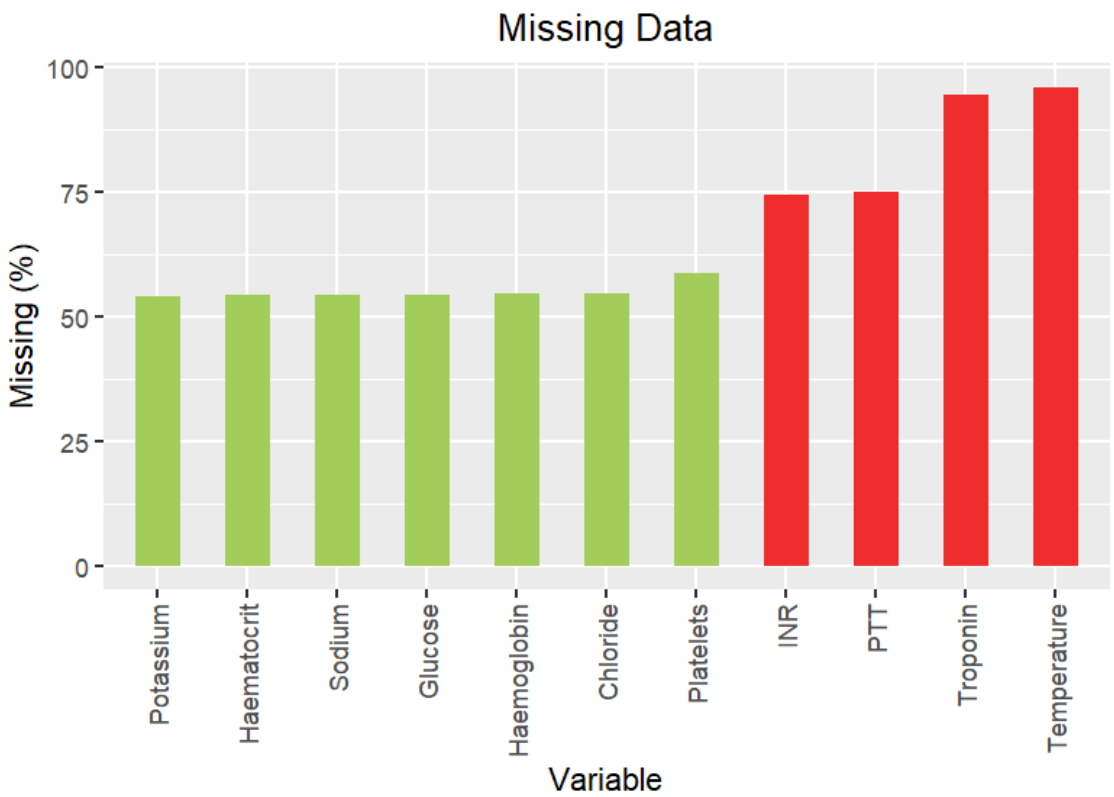
Characteristic	N = 2303
Number of previous ICU stays (within hospital admission)	
0	2,115 (92.0%)
1	171 (7.4%)
2	13 (0.6%)
3	3 (0.1%)
4	1 (<0.1%)
Age (years)	
18-40	14 (0.6%)
41-50	93 (4.0%)
51-60	386 (16.8%)
61-70	769 (33.4%)
71-80	668 (29.0%)
81-90	318 (13.8%)
91+	55 (2.4%)

Characteristic	N = 2303
Sex	
Male	1,237 (54%)
Female	1,066 (46%)
Ethnicity	
White	1,671 (72.6%)
Black	217 (9.4%)
Asian	122 (5.3%)
Hispanic/Latino	49 (2.1%)
Other	66 (2.9%)
Missing	178 (7.7%)
Insurance	
Medicare	1,418 (61.6%)
Private	537 (23.3%)
Medicaid	288 (12.5%)
Other	44 (1.9%)
Missing	16 (0.7%)
Admission Type	
Emergency	1,278 (55.5%)
Observation	365 (15.9%)
Urgent	335 (14.5%)
Elective	325 (14.1%)
Platelets (k/µL)	
Mean (SD)	265.8 (133.1)
N missing (%)	1,347 (58.5)
Glucose (mg/dL)	
Mean (SD)	137.2 (78.9)
N missing (%)	1,252 (54.4)
Chloride (mEq/L)	
Mean (SD)	100.4 (8.0)
N missing (%)	1,259 (54.7)
Potassium (mEq/L)	
Mean (SD)	4.7 (4.9)
N missing	1,244 (54.0)

Characteristic	N = 2303
Haemoglobin (g/dL)	
Mean (SD)	10.6 (2.6)
N missing (%)	1,258 (54.6)
Haematocrit (%)	
Mean (SD)	32.7 (6.7)
N missing (%)	1,249 (54.2)
Sodium (mEq/L)	
Mean (SD)	135.3 (13.0)
N missing (%)	1,250 (54.3)
Drugs taken	
Ondansetron	397 (17.2%)
Calcium Gluconate	926 (40.2%)
Death in ICU	
Survived	1775 (77.1%)
Deceased	528 (22.9%)
Length of stay (days)	
Mean (SD)	3.9 (5.2)

Table 4: Caption

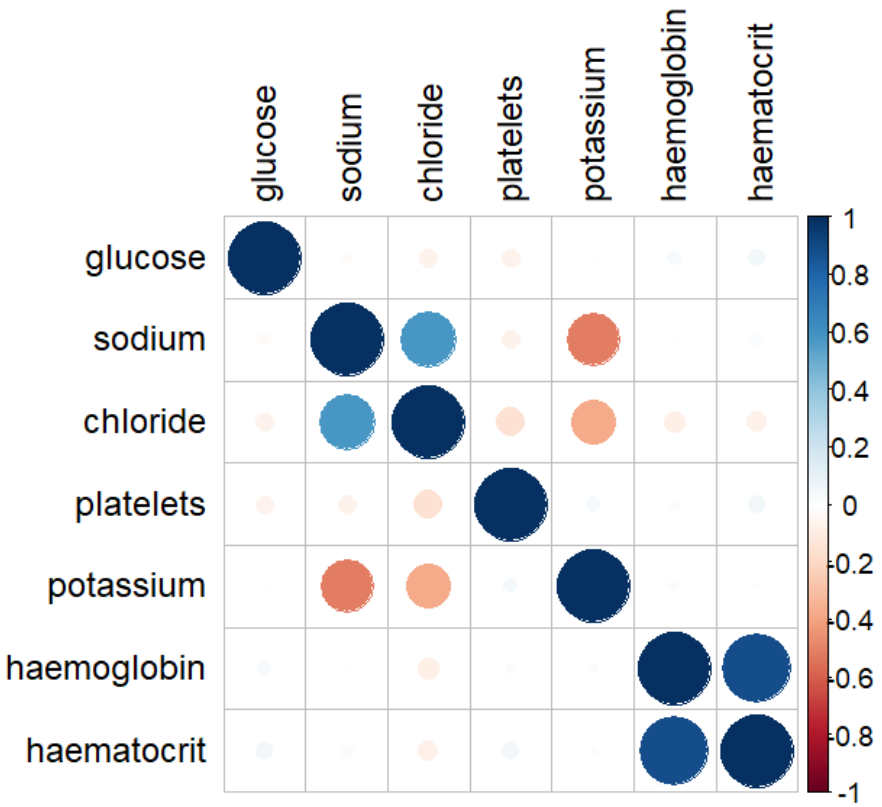
3.2 Missing Data



All of the data variables shown in this graph have highly significant amounts of missing data with not one variable showing a percentage of missing data less than 50 %Ṫhis is not very conducive for the inclusion of these variables in the prediction models since when data is more than 40 % missing it can only be considered as hypothesis generating and therefore will likely not provide accurate predictions. However, some data must be used so for the purposes of the continuation of the study the variables in green are kept as predictors in the model.

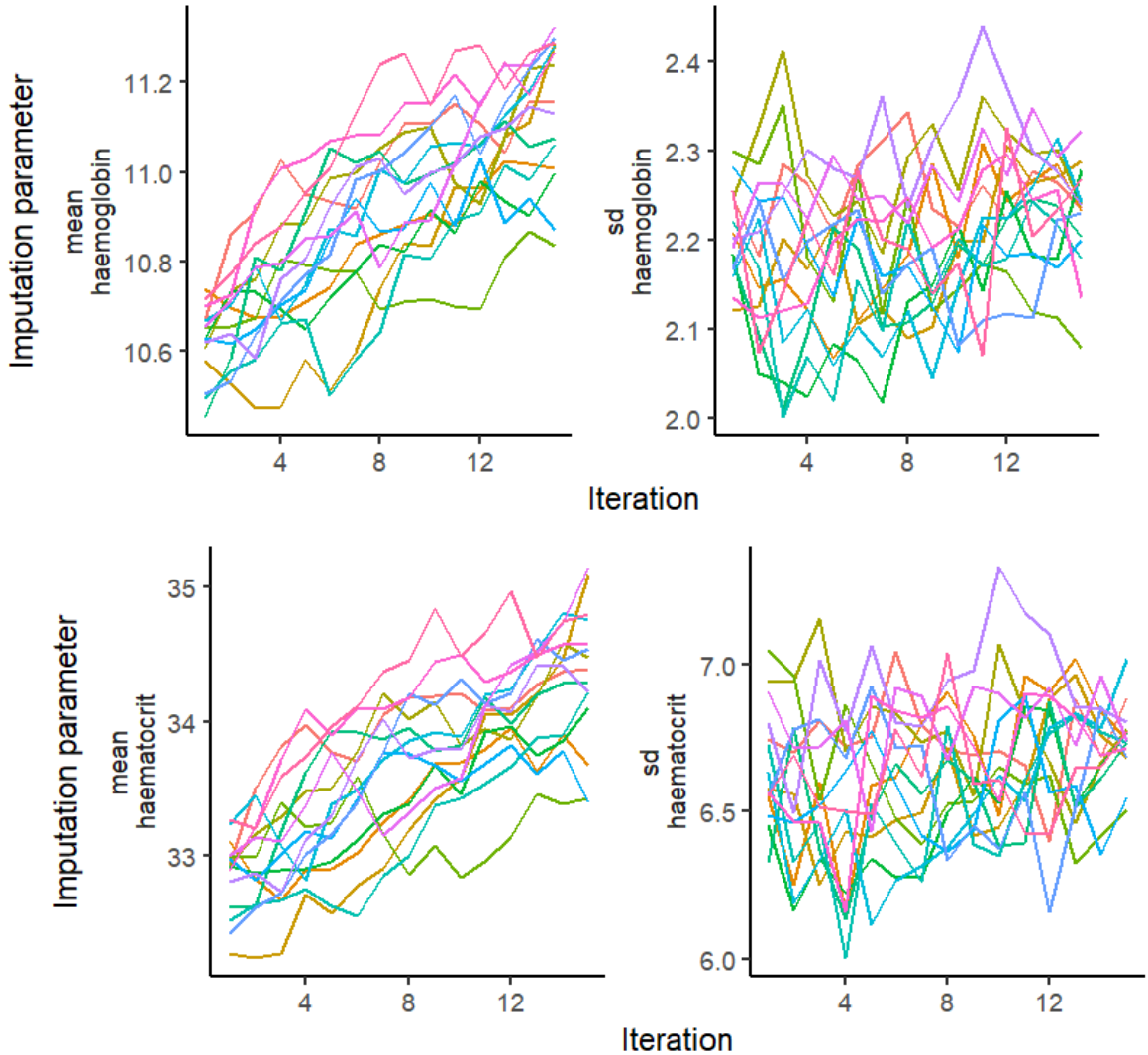
3.3 Imputation

3.3.1 Correlation Plot

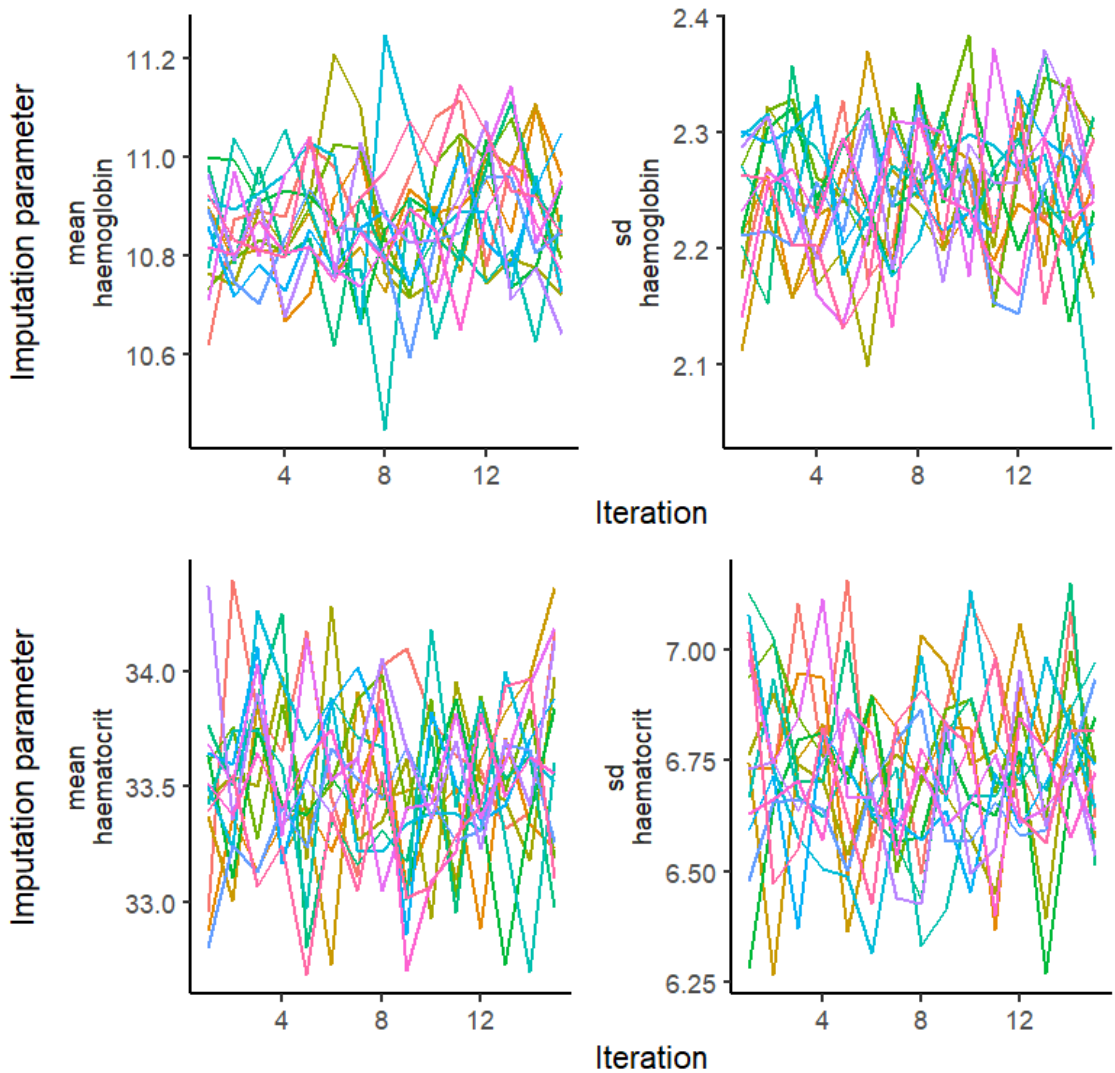


The correlation graph shows that most of the variables are not significantly correlated enough that it would affect the imputation process by not enabling the model to converge for all variables. However the haematocrit and the haemoglobin variables are very significantly correlated, and therefore they have high levels of multicolinearity, which should hinder the models potential to converge and produce reliable imputed data. An adjustment to the prediction matrix is needed to remove the relationship between haematocrit and haemoglobin from the imputation process.

3.3.2 Trace Plots



Clearly shown in the trace plots above the mean of the data for both the haemoglobin and the haematocrit variables tends to increase with each imputation iteration. Ideally there should be a point where the mean should remain mostly unchangable from iteration to iteration. To fix this the predictor matrix should be amended. The way to do this is to not have the haematocrit be influential on the imputation of the haemoglobin and vice versa. The predictor matrix determines this with the use of binary inputs. The default is a matrix full of 1s which allows for an imputation where every variable affects the imputation of each other. By changing the 1s that refer to the haematocrit/haemoglobin interaction to 0 the trace plots should be corrected.



These plots show a more correct imputation where the mean reaches a point where it remains mostly unchangable from iteration to iteration.

3.3.3 Categorical Missing Values

3.4 Model Performance

The linear regression model is the most successful according to the MAE metric with just over a 0.1 hour difference between the linear regression's MAE of 2.79 to the other three that are all either 2.9 or 2.91. Where this is the case for the MAE the inverse is true for both the RMSE and the R^2 values. In this case linear regression was worst performing model by a significant amount. Random forest beat out the two penalisation methods and would therefore be considered the superior model based on these metrics alone.

Model	RMSE	MAE	R^2
Linear Regression	5.69	2.79	-0.24
LASSO	5.05	2.90	0.03
Ridge	5.05	2.90	0.03
Random Forest	4.99	2.91	0.04

Table 5: Caption

4 Discussion

4.1 Findings

Out of the four prediction models random forest was found to be the most reliable

4.2 Comparison to the Literature

Less accurate findings than the literature found although there is a higher possibility that more significant findings will be published, other paper used more data variables and had less missing data, less prior literature used the MIMIC-IV database, similar result was found in terms of which machine learning method was most effective/suitable

4.3 The Effect of Missing Data

missing data levels were high resulting in an inaccurate and less clinically and statistically significant result, imputation methods only work well when missing data proportion is minimal, complete case analysis would not be feasible as the sample size would be below the required amount

4.4 Limitations

A lot of missing data, not taking into account of those who died during their stay, only a small amount of variables, limited prior study on the specific topic

4.5 Future Recommendations

larger datasets, assess on data with less missing data or a significant sample size of complete case data, investigate new variables, take into account the patients who die during stay, explore new algorithms, test on new data/demographics

5 Conclusion

Findings were not clinincally significant enough for this model to be trusted to make accurate predictions on lung cancer patients length of stay in the ICU but due to the depths of its limitations the possibility for an effective prediction model such as this to be created cannot be ruled out. Further study with fewer limitations may provide a more reliable prediction model

References

[1] Marzyeh Ghassemi et al. *A Review of Challenges and Opportunities in Machine Learning for Health*. Tech. rep.

[2] Jennie Martin. *SQL code to load MIMIC-IV schema*. URL: https://pages.doit.wisc.edu/JLMARTIN22/mimic-code/-/blob/sqlfluff/mimic-iv/buildmimic/postgres/create.sql.

[3] International Statistical Classification of Diseases and Related Health Problems. *ICD-10 Version:2019*. 2019. URL: https://icd.who.int/browse10/2019/en.

[4] Md Kamrul Hasan et al. *Missing value imputation affects the performance of machine learning: A review and analysis of the literature (2010–2021)*. Jan. 2021. DOI: 10.1016/j.imu.2021.100799.

[5] Minqian Yan et al. "Comparison of different approaches in handling missing data in longitudinal multiple-item patient-reported outcomes: a simulation study". In: *Health and Quality of Life Outcomes* 23 (1 Dec. 2025). ISSN: 14777525. DOI: 10.1186/s12955-025-02364-0.

[6] Jiaxu Peng, Jungpil Hahn, and Ke Wei Huang. "Handling Missing Values in Information Systems Research: A Review of Methods and Assumptions". In: *Information Systems Research* 34 (1 Mar. 2023), pp. 5–26. ISSN: 1526-5536. DOI: 10.1287/isre.2022.1104.

[7] Joseph R. Dettori, Daniel C. Norvell, and Jens R. Chapman. “The Sin of Missing Data: Is All Forgiven by Way of Imputation?” In: *Global Spine Journal* 8 (8 Dec. 2018), pp. 892–894. ISSN: 21925690. DOI: 10.1177/2192568218811922.

[8] Rebecca R. Andridge and Roderick J.A. Little. *A review of hot deck imputation for survey non-response*. Apr. 2010. DOI: 10.1111/j.1751-5823.2010.00103.x.

[9] Stylianos Georgiadis et al. “Challenge of missing data in observational studies: investigating cross-sectional imputation methods for assessing disease activity in axial spondyloarthritis”. In: *RMD Open* 11 (1 Feb. 2025). ISSN: 20565933. DOI: 10.1136/rmdopen-2024-004844.

[10] Ian R White, Patrick Royston, and Angela M Wood. “Tutorial in Biostatistics Multiple imputation using chained equations: Issues and guidance for practice”. In: (2010). DOI: 10.1002/sim.4067. URL: https://onlinelibrary.wiley.com/doi/10.1002/sim.4067.

[11] Jacques Emmanuel Galimard et al. “Heckman imputation models for binary or continuous MNAR outcomes and MAR predictors”. In: *BMC Medical Research Methodology* 18 (1 Aug. 2018). ISSN: 14712288. DOI: 10.1186/s12874-018-0547-1.

[12] Tim P Morris, Ian R White, and Patrick Royston. *Tuning multiple imputation by predictive mean matching and local residual draws*. Tech. rep. 2014, p. 75. URL: http://www.biomedcentral.com/1471-2288/14/75.

[13] Muhammad Arbab Arshad, Sakib Shahriar, and Khizar Anjum. “The Power Of Simplicity: Why Simple Linear Models Outperform Complex Machine Learning Techniques – Case Of Breast Cancer Diagnosis”. In: (June 2023). URL: http://arxiv.org/abs/2306.02449.

[14] Dastan Maulud and Adnan M. Abdulazeez. “A Review on Linear Regression Comprehensive in Machine Learning”. In: *Journal of Applied Science and Technology Trends* 1 (2 Dec. 2020), pp. 140–147. DOI: 10.38094/jastt1457.

[15] Muhammad Ibrahim. “Evolution of Random Forest from Decision Tree and Bagging: A Bias-Variance Perspective”. In: *Dhaka University Journal of Applied Science and Engineering* 7 (1 Feb. 2023), pp. 66–71. ISSN: 2218-7413. DOI: 10.3329/dujase.v7i1.62888.

[16] Jitendra Singh Kushwah et al. “Comparative study of regressor and classifier with decision tree using modern tools”. In: *Materials Today: Proceedings* 56 (Jan. 2022), pp. 3571–3576. ISSN: 2214-7853. DOI: 10.1016/j.matpr.2021.11.635.

[17] Hemant Ishwaran. “The Effect of Splitting on Random Forests”. In: *Machine Learning* 99 (1 Apr. 2015), pp. 75–118. ISSN: 1573-0565. DOI: 10.1007/s10994-014-5451-2.

[18] John Kitchin and Siddhant Lambor. *split random forest*. 2020. URL: https://kitchingroup.cheme.cmu.edu/s20-06681/09-random-forests/09-random-forests.html?.

[19] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics.

[20] Gérard Biau and Erwan Scornet. “A Random Forest Guided Tour”. In: (Nov. 2015). URL: http://arxiv.org/abs/1511.05741.

[21] Leo Breiman. “Random Forests”. In: (Jan. 2001).

[22] Gaoxia Jiang and Wenjian Wang. “Error estimation based on variance analysis of k-fold cross-validation”. In: *Pattern Recognition* 69 (Sept. 2017), pp. 94–106. ISSN: 0031-3203. DOI: 10.1016/j.patcog.2017.03.025.

[23] Juan D. Rodríguez, Aritz Pérez, and Jose A. Lozano. “A general framework for the statistical analysis of the sources of variance for classification error estimators”. In: *Pattern Recognition* 46 (3 Mar. 2013), pp. 855–864. ISSN: 00313203. DOI: 10.1016/j.patcog.2012.09.007.

[24] Sudhir Varma and Richard Simon. “Bias in error estimation when using cross-validation for model selection”. In: *BMC Bioinformatics* 7 (Feb. 2006). ISSN: 14712105. DOI: 10.1186/1471-2105-7-91.

[25] Stephen Bates, Trevor Hastie, and Robert Tibshirani. “Cross-validation: what does it estimate and how well does it do it?” In: (July 2022). DOI: 10.1080/01621459.2023.2197686. URL: http://arxiv.org/abs/2104.00673%20http://dx.doi.org/10.1080/01621459.2023.2197686.

[26] Damjan Krstajic et al. “Cross-validation pitfalls when selecting and assessing regression and classification models”. In: *Journal of Cheminformatics* 6 (1 Mar. 2014). ISSN: 17582946. DOI: 10.1186/1758-2946-6-10. URL: https://www.machinelearningmastery.com/nested-cross-validation-for-machine-learning-with-python/%20https://machinelearningmastery.com/nested-cross-validation-for-machine-learning-with-python/.

[27] *On Leakage in Machine Learning Pipelines*. Tech. rep. URL: https://hunch.net/?p=22.

Appendix